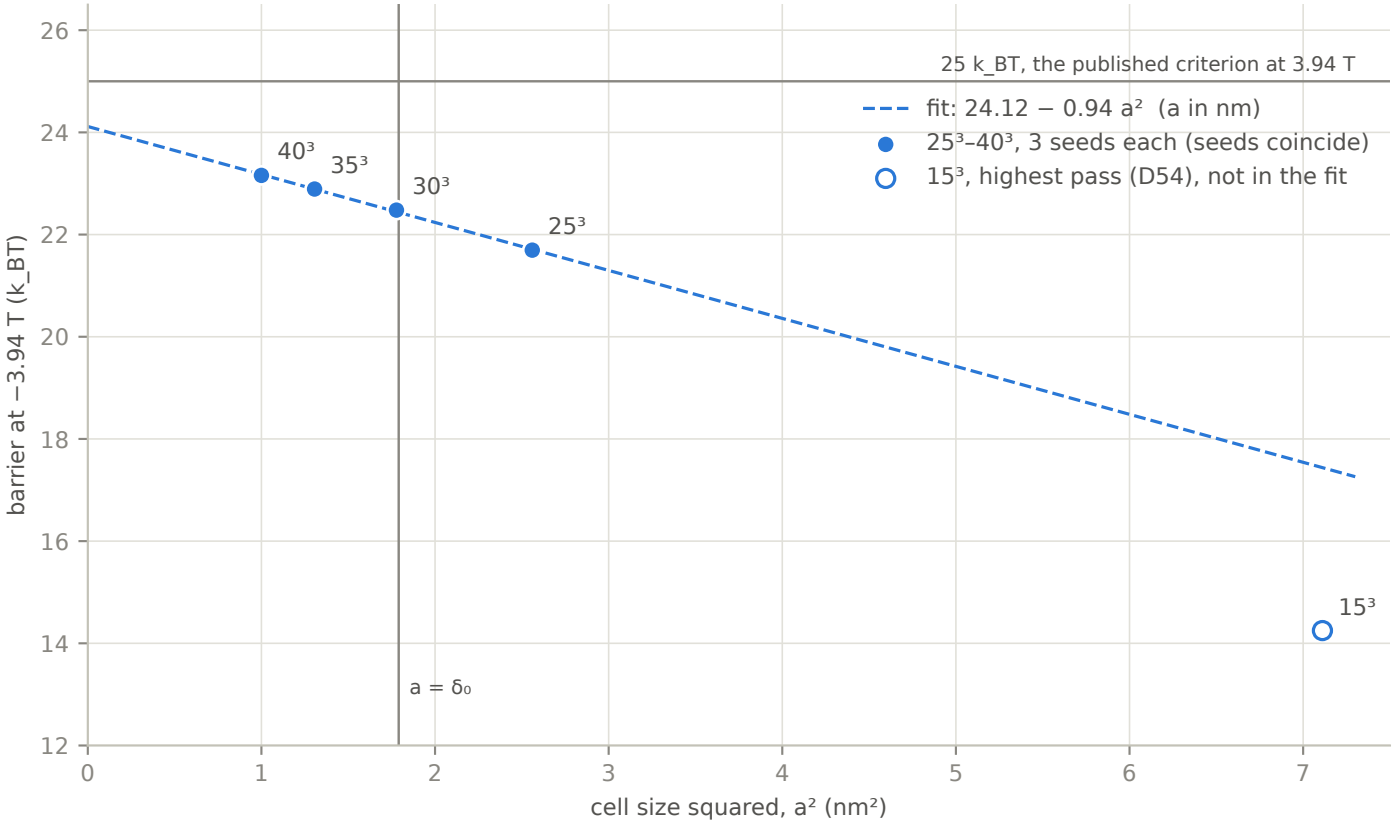
Energy barrier against cell size

From 25³ to 40³ the landscape is one clean pass: all 12 chains converge, seeds agree to four decimals, and all 96 Newton starts land on that pass or the start. The barrier still rises as the mesh is refined, from 21.69 to 23.16 k_BT, and extrapolates to about 24.1 k_BT. A cell of δ₀ reads about 7.0% low, and error scales as a². The nucleus has the same volume on every mesh, about 17 nm³, which is less than one 15³ cell (19 nm³).



mesh	a (nm)	a / δ₀	converged	iterations	barrier	vs E₀	mode (T)	reversed	nm³
25³	1.600	1.196	3 of 3	544–1018	21.6949	-10.0%	-1.500	4	16.4
30³	1.333	0.996	3 of 3	603–748	22.4783	-6.8%	-1.644	7	16.6
35³	1.143	0.854	3 of 3	614–807	22.8910	-5.1%	-1.641	11	16.4
40³	1.000	0.747	3 of 3	818–936	23.1582	-4.0%	-1.653	17	17.0
15³ (D54)	2.667	1.993	0 of 3 at 1e-4	—	14.2492	-40.9%	−3.510	0–1	one cell is 19

Barrier in k_BT. Mode is the lowest curvature; reversed and nm³ describe the nucleus at the saddle. Fit through all four meshes: E₀ = 24.12 k_BT; through the three finest: 24.03 k_BT. Pushes reached start and end on 74 of 86 saddles, and both sides reached the end on the others.

The 40 nm $Nd_2Fe_{14}B$ cube of D32: $\mu_0 M_s$ = 1.61 T, K = 4.3 MJ/m³, A = 7.7 pJ/m, so δ₀ = 1.34 nm. Field −3.94 T, 1° off z, 10 images, torque 1e-4, limit 2000 iterations, 3 seeds, L-BFGS as in gneb(). Newton from all 8 inner images of each chain, curvatures by Lanczos, pushes of 0.03 rad. Reversed cells are those with m·m_start < 0 at the saddle, so the volume is a rough measure at about ±½ cell. The fit is E₀ + c a² through the four meshes, and E₀ is an extrapolation below the finest cell run. The 15³ point is from D54 and is not in the fit. The 25 k_BT line is the published criterion at this field (Fischbacher et al. 2017), from finite elements at 1.5 nm. Meshes 16–24 were not run, so where the staircase starts is unknown. Decision D56. Regenerate with benchmarks/barrier_cell_size.py.